



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab Programs

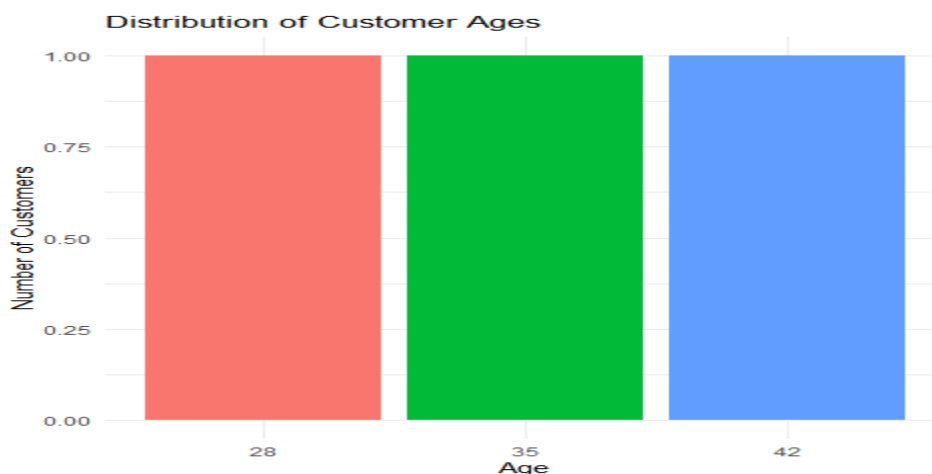
SET 7- Customer Demographics Analysis

7A. Bar Chart

Program:

```
customer_data <- data.frame(  
  CustomerID=c(1,2,3),  
  Age=c(28,35,42),  
  Gender=c("Female","Male","Female"),  
  Income=c(50000,60000,75000))  
  
ggplot(customer_data,  
  aes(x=factor(Age),  
    fill=factor(Age)))+  
  geom_bar()+  
  labs(title="Distribution of Customer Ages",  
    x="Age",  
    y="Number of Customers")+  
  theme_minimal()+  
  theme(legend.position="none")
```

Output:



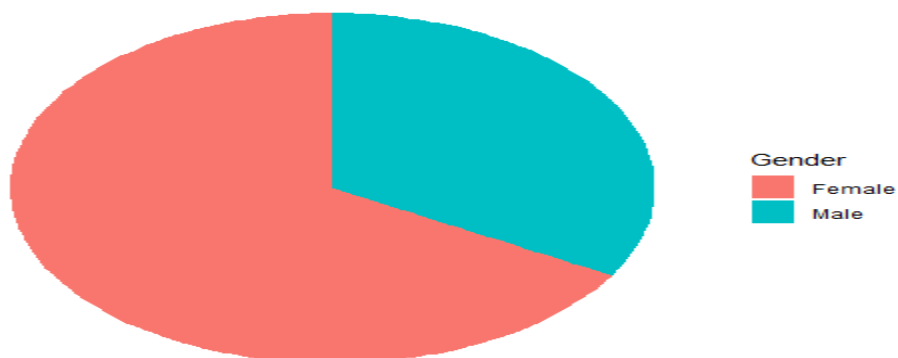
7B. Pie Chart

Program:

```
gender_data <- data.frame(table(customer_data$Gender))  
colnames(gender_data) <- c("Gender","Count")  
ggplot(gender_data,  
aes(x="",y=Count,fill=Gender))+  
geom_bar(stat="identity",width=1)+  
coord_polar("y")+  
labs(title="Customer Distribution by Gender")+  
theme_void()
```

Output:

Customer Distribution by Gender



7C. Table

Program:

```
kable(  
customer_data,  
caption="Customer Demographics Information")
```

Output:

```
Table: Customer Demographics Information  
| CustomerID| Age|Gender | Income|  
|-----:|---:|:-----|-----:|  
|          1| 28|Female | 50000|  
|          2| 35|Male   | 60000|  
|          3| 42|Female | 75000|  
> |
```

7D. Combined Interactive Dashboard

Program:

```
library(plotly)

bar_plot <- ggplotly(
  ggplot(customer_data,
    aes(x=factor(Age), fill=factor(Age))) +
  geom_bar() +
  labs(title="Customer Age Distribution") +
  theme_minimal() +
  theme(legend.position="none"))

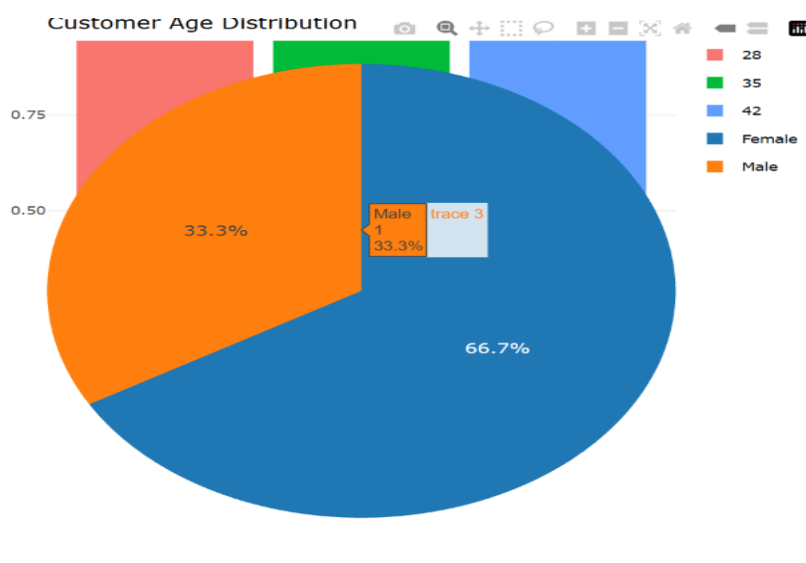
gender_data <- data.frame(
  Gender=c("Female","Male"),
  Count=c(2,1))

pie_plot <- plot_ly(
  gender_data,
  labels=~Gender,
  values=~Count,
  type="pie")

dashboard <- subplot(bar_plot, pie_plot, nrows=2)

dashboard
```

Output:



SET 8- Employee Performance Analysis

8A. Line Chart

Step 1: Install Packages (Run Only Once)

```
install.packages("ggplot2")
```

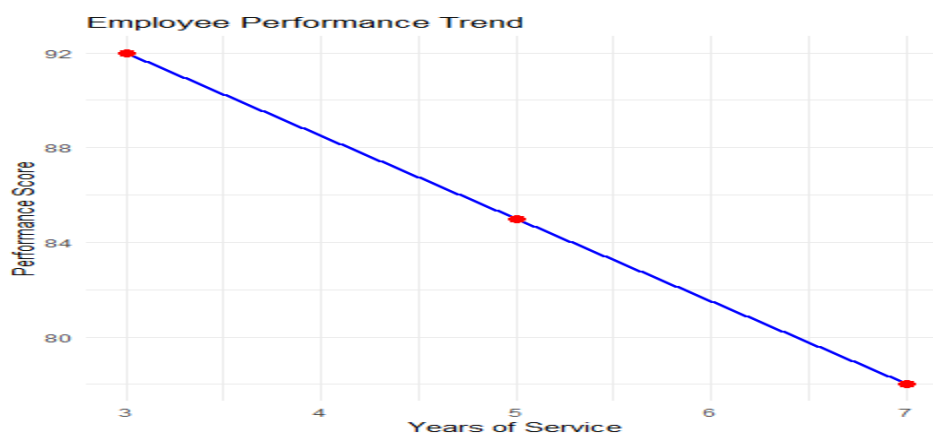
```
install.packages("plotly")
```

```
install.packages("knitr")
```

Program:

```
employee_data <- data.frame(  
  EmployeeID=c(1,2,3),  
  Department=c("Sales","HR","Marketing"),  
  YearsOfService=c(5,3,7),  
  PerformanceScore=c(85,92,78))  
  
ggplot(employee_data,  
  aes(x=YearsOfService,  
  y=PerformanceScore,  
  group=1))+  
  geom_line(color="blue",size=1)+  
  geom_point(color="red",size=3)+  
  labs(title="Employee Performance Trend",  
  x="Years of Service",  
  y="Performance Score")+  
  theme_minimal()
```

Output:

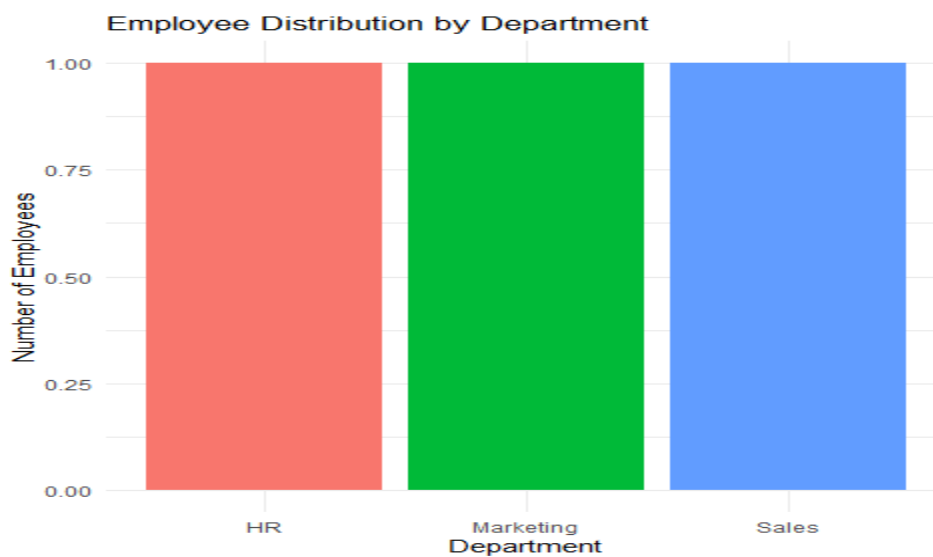


8B. Bar Chart

Program:

```
ggplot(employee_data,  
aes(x=Department,  
fill=Department))+  
geom_bar()+  
labs(title="Employee Distribution by Department",  
x="Department",  
y="Number of Employees")+  
theme_minimal()+  
theme(legend.position="none")
```

Output:



8C. Table

Program:

```
kable(  
employee_data,  
caption="Employee Performance Data")
```

Output:

Table: Employee Performance Data

EmployeeID	Department	YearsOfService	PerformanceScore
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

> |

8D. Interactive Dashboard

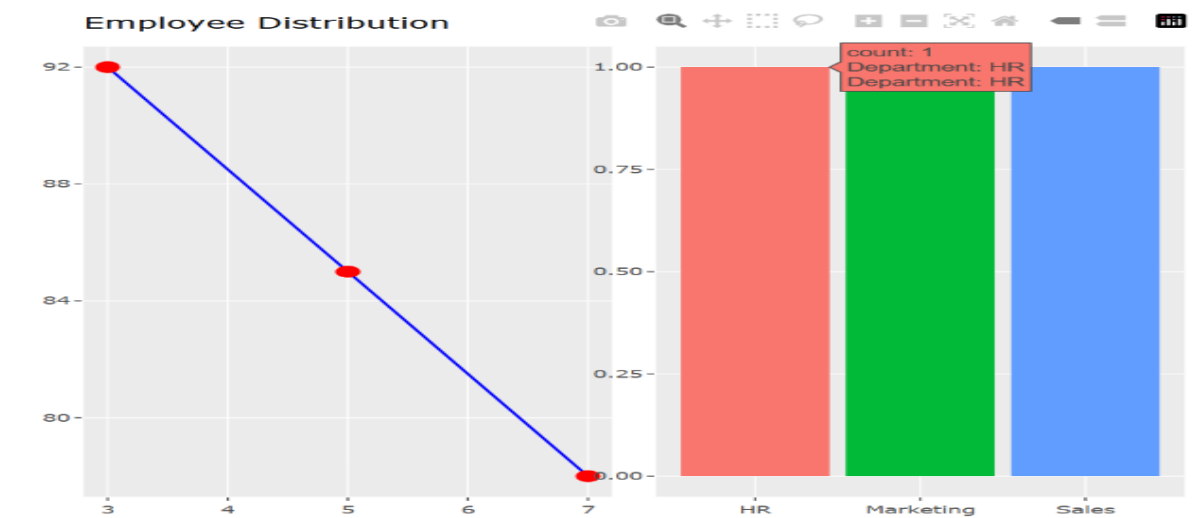
Program:

```
library(plotly)

line_plot <- ggplotly(
  ggplot(employee_data,
    aes(x=YearsOfService,
      y=PerformanceScore, group=1))+
  geom_line(color="blue")+
  geom_point(color="red",size=3)+
  labs(title="Employee Performance Trend"))

bar_plot <- ggplotly(
  ggplot(employee_data,
    aes(x=Department,
      fill=Department))+
  geom_bar()+
  labs(title="Employee Distribution")+
  theme(legend.position="none")) dashboard <- subplot(
  line_plot,bar_plot, nrows=1,
  shareX=FALSE,
  shareY=FALSE)
dashboard
```

Output:



Set 9- Online_Learning_Activity.csv

9A. Import Dataset, Histogram & Boxplot

Program:

```
data <- read.csv(file.choose())
```

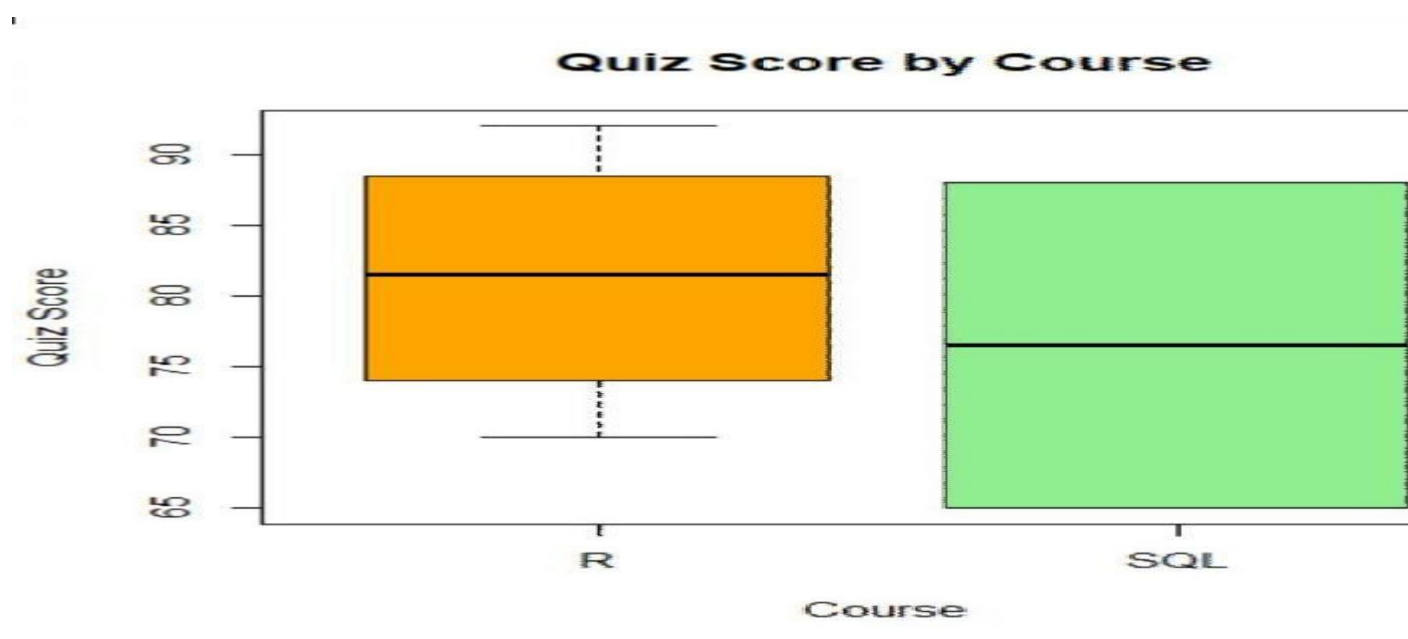
Histogram:

```
ggplot(data,aes(x=Quiz_Score))+  
geom_histogram(binwidth=5,fill="skyblue",color="black")+  
labs(title="Distribution of Quiz Scores",  
x="Quiz Score", y="Number of Students")+  
theme_minimal()
```

Boxplot:

```
ggplot(data,aes(x=Course,  
y=Quiz_Score, fill=Course))+  
geom_boxplot()+  
labs(title="Quiz Score by Course",  
x="Course",  
y="Quiz Score")+  
theme_minimal()
```

Output:

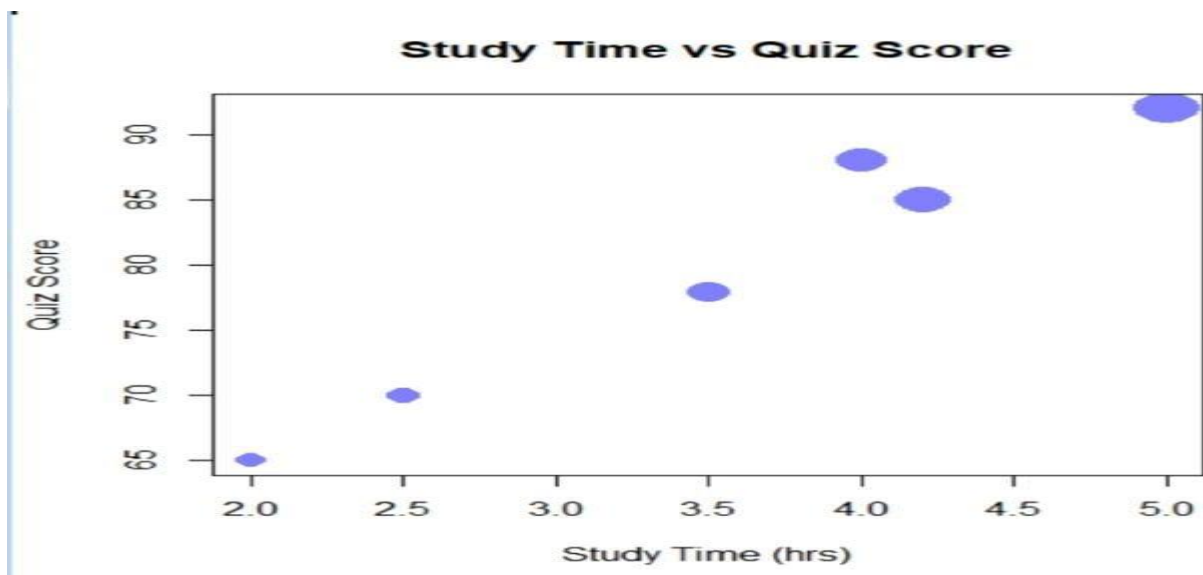


9B. Scatter Plot with Bubble Size

Program:

```
ggplot(data,  
aes(x=Study_Time,  
y=Quiz_Score,  
size=Videos_Watched,  
color=Course))+  
geom_point(alpha=0.6)+  
labs(title="Study Time vs Quiz Score",  
x="Study Time (hrs)",  
y="Quiz Score",  
size="Videos Watched")+  
theme_minimal()
```

Output:



9C. Date Conversion, Monthly Average & Line Chart

Program:

Convert Date

```
data$Login_Date <- as.Date(data$Login_Date)
```

Extract Month

```
data$Month <- format(data$Login_Date,"%Y-%m")
```

Average Quiz Score Per Month


```
monthly <- data %>%
group_by(Month) %>%
summarise(Avg_Quiz=mean(Quiz_Score))
```

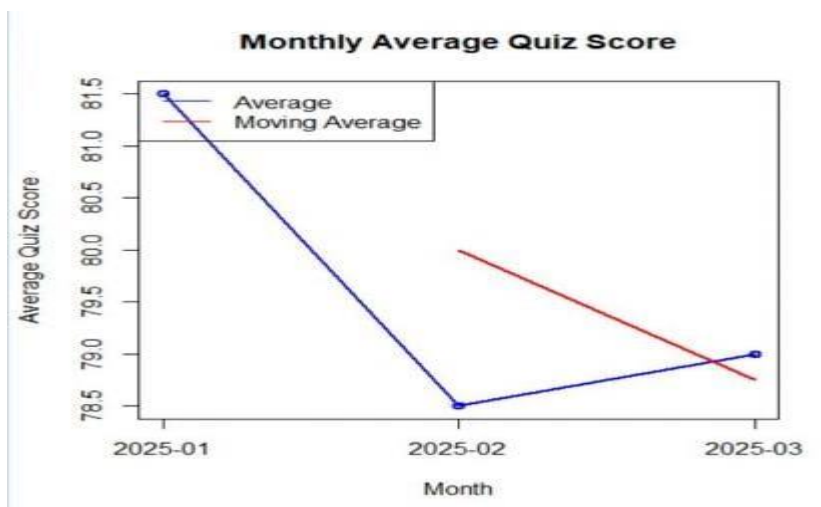
Moving Average

```
monthly$MovingAvg <- rollmean(
monthly$Avg_Quiz,
k=2,fill=NA,
align="right")
```

Line Chart

```
ggplot(monthly,aes(x=Month,y=Avg_Quiz,group=1))+
geom_line(color="blue",size=1)+
geom_point(size=3,color="red")+
geom_line(aes(y=MovingAvg),
color="green",
linetype="dashed",
size=1)+
labs(title="Average Monthly Quiz Score",
x="Month",
y="Average Quiz Score")+
theme_minimal()
```

Output:



9D. Interactive Dashboard

Program:

```
library(shiny)

data <- data.frame(
  Student_ID=c("L01","L02","L03","L04","L05","L06"),
  Gender=c("Male","Female","Male","Female","Male","Female"),
  Course=c("R","R","SQL","R","R","SQL"),
  Study_Time=c(3.5,4.2,2.0,5.0,2.5,4.0),
  Quiz_Score=c(78,85,65,92,70,88)
)

ui <- fluidPage(
  titlePanel("Online Learning Dashboard"),
  sidebarLayout(
    sidebarPanel(
      selectInput("gender",
        "Gender",
        choices=c("All",unique(data$Gender))),
      selectInput("course",
        "Course",
        choices=c("All",unique(data$Course))) ),
    mainPanel(
      fluidRow(
        column(6,
          plotOutput("bar")),
        column(6,
          plotOutput("scatter"))
      ) ) )
)

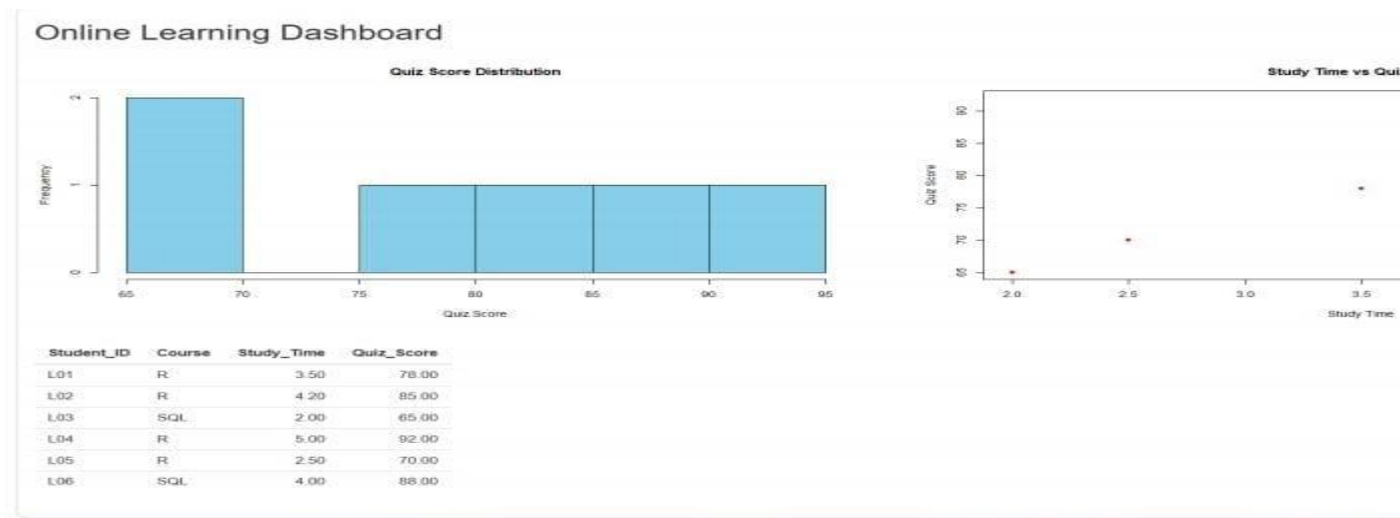
server <- function(input,output){
  output$bar <- renderPlot({
```

```

d <- data
if(input$gender!="All")
d <- subset(d,Gender==input$gender)
if(input$course!="All")
d <- subset(d,Course==input$course)
avg <- tapply(d$Quiz_Score,d$Course,mean)
barplot(avg,
col="skyblue",
main="Average Quiz Score by Course",
ylab="Quiz Score")
})
output$scatter <- renderPlot({
d <- data
if(input$gender!="All")
d <- subset(d,Gender==input$gender)
if(input$course!="All")
d <- subset(d,Course==input$course)
plot(d$Study_Time,
d$Quiz_Score,
pch=16,
col="red",
xlab="Study Time",
ylab="Quiz Score",
main="Study Time vs Quiz Score")
})
}
shinyApp(ui,server)

```

Output:



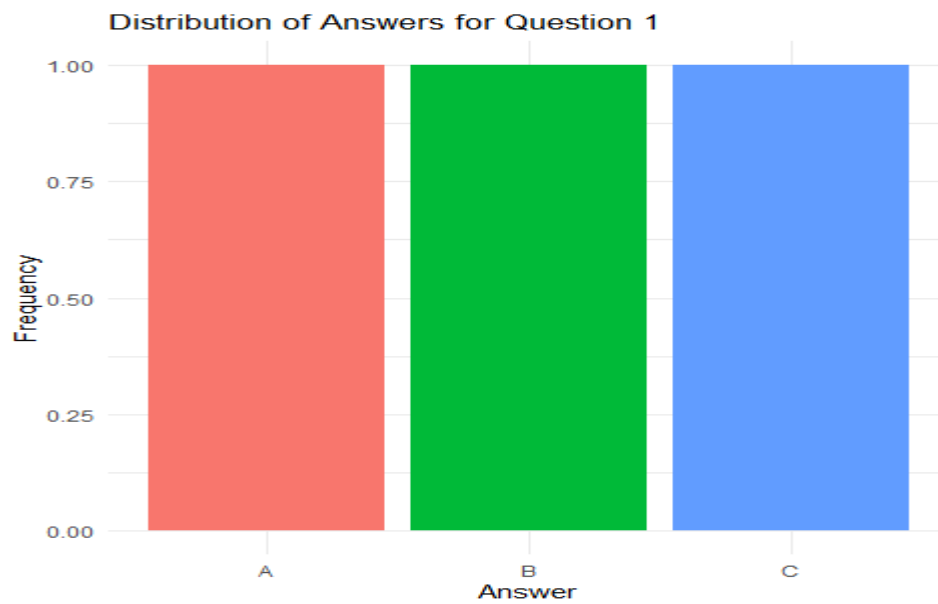
Set 10: Survey Responses Analysis

10A. Grouped Bar Chart

Program:

```
survey_data <- data.frame(  
  SurveyID=c(1,2,3),  
  Question1=c("A","B","C"),  
  Question2=c("B","A","A"),  
  Question3=c("C","D","B"))  
  
q1_count <- as.data.frame(table(survey_data$Question1))  
colnames(q1_count) <- c("Answer","Count")  
  
ggplot(q1_count,  
  aes(x=Answer,  
    y=Count,  
    fill=Answer))+  
  geom_bar(stat="identity")+  
  labs(title="Distribution of Answers for Question 1",  
    x="Answer",  
    y="Frequency")+  
  theme_minimal()+  
  theme(legend.position="none")
```

Output:

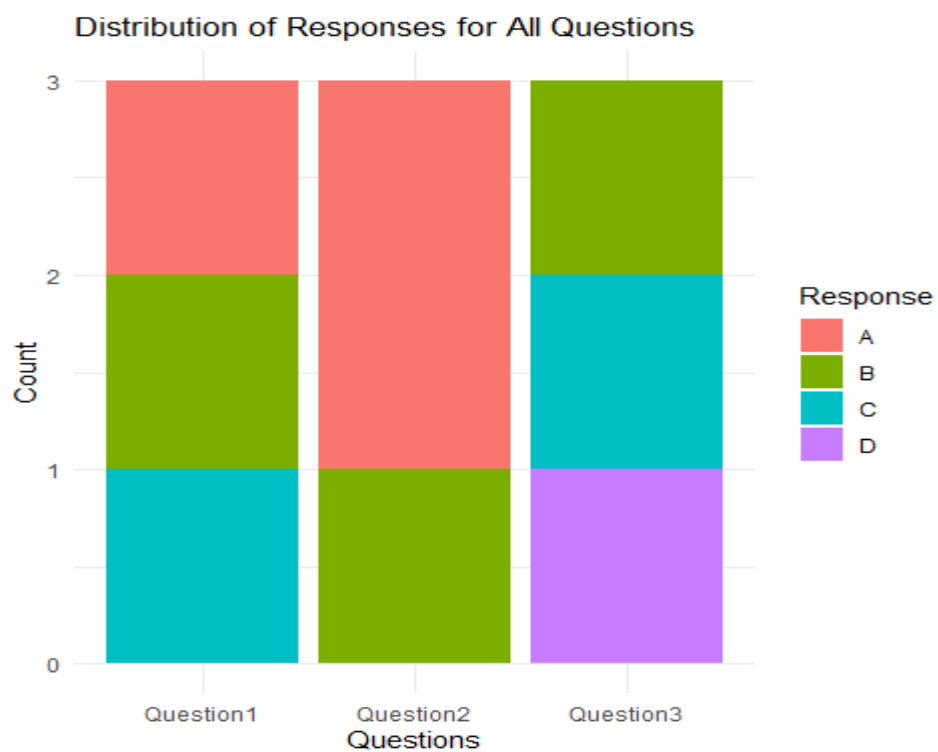


10 B. Stacked Bar Chart

Program:

```
survey_long <- melt(
  survey_data,
  id.vars="SurveyID",
  variable.name="Question",
  value.name="Response")
ggplot(survey_long,
  aes(x=Question,
  fill=Response))+
  geom_bar()+
  labs(title="Distribution of Responses for All Questions",
  x="Questions",
  y="Count",
  fill="Response")+
  theme_minimal()
```

Output:



10C. Table

Program:

```
kable(  
survey_data,  
caption="Survey Response Data"  
)
```

Output:

Table: Survey Response Data

SurveyID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

10D. Interactive Dashboard (R)

Program:

```
bar_plot <- ggplotly(  
ggplot(q1_count,
```

```

aes(x=Answer,
y=Count,
fill=Answer))+
geom_bar(stat="identity")+
labs(title="Question 1 Responses")+
theme_minimal()+
theme(legend.position="none"))

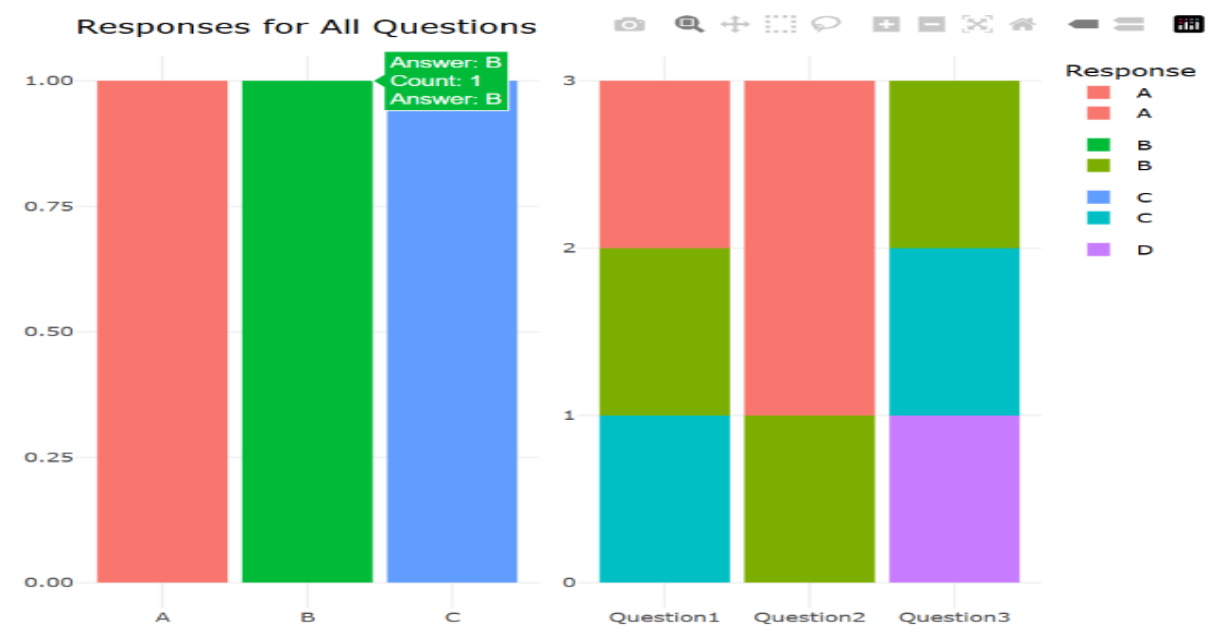
stacked_plot <- ggplotly(
ggplot(survey_long,
aes(x=Question,
fill=Response))+
geom_bar()+
labs(title="Responses for All Questions")+
theme_minimal())

dashboard <- subplot(bar_plot,
stacked_plot, nrows=1,
shareX=FALSE,
shareY=FALSE)

dashboard

```

Output:



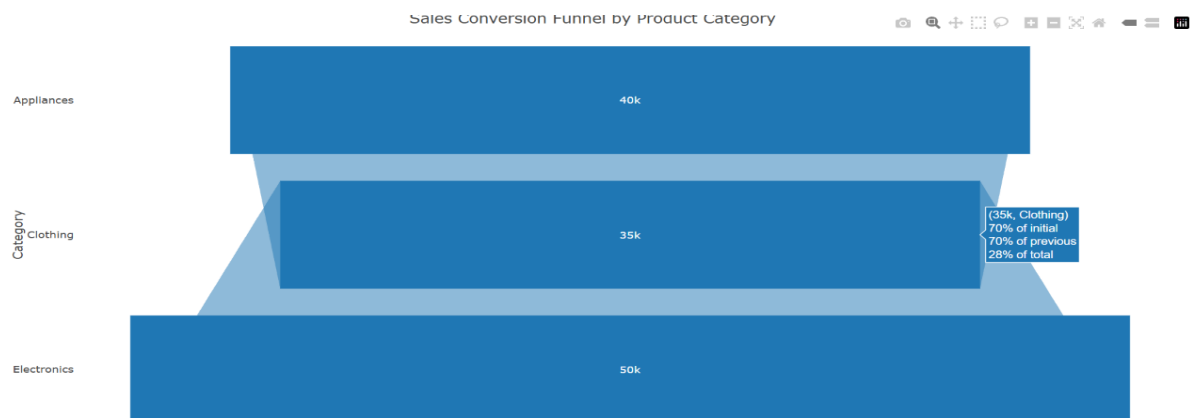
SET 11- Product Category Analysis

11 A. Funnel Chart

Program:

```
product_data <- data.frame(  
  Category=c("Electronics","Clothing","Appliances"),  
  Sales=c(50000,35000,40000))  
  
plot_ly( product_data,  
  type="funnel",  
  y=~Category,  
  x=~Sales) %>%  
  layout(title="Sales Conversion Funnel by Product Category")
```

Output:



11 B. Table

Program:

```
kable(  
  product_data,  
  caption="Product Category Sales")
```

Output:

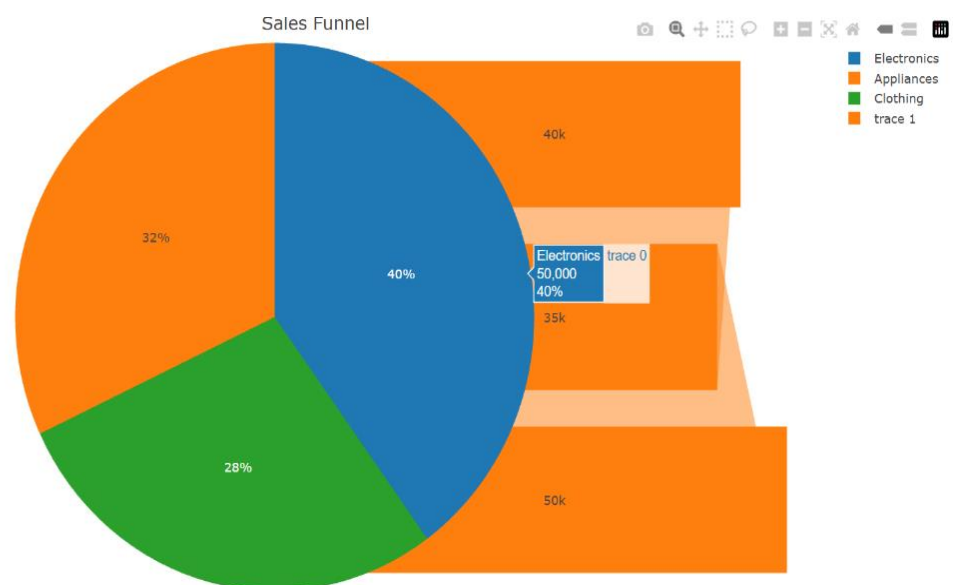
```
Table: Product Category Sales  
  
|Category| Sales|  
|:-----|:-----|  
|Electronics| 50000|  
|Clothing| 35000|  
|Appliances| 40000|  
> |
```


11 C. Interactive Dashboard (R)

Program:

```
pie_plot <- plot_ly(  
  product_data,  
  labels=~Category,  
  values=~Sales,  
  type="pie") %>%  
  layout(title="Sales Distribution")  
  
funnel_plot <- plot_ly(  
  product_data,  
  type="funnel",  
  y=~Category,  
  x=~Sales) %>%  
  layout(title="Sales Funnel")  
  
dashboard <- subplot(  
  pie_plot,  
  funnel_plot,  
  nrows=1)  
dashboard
```

Output:



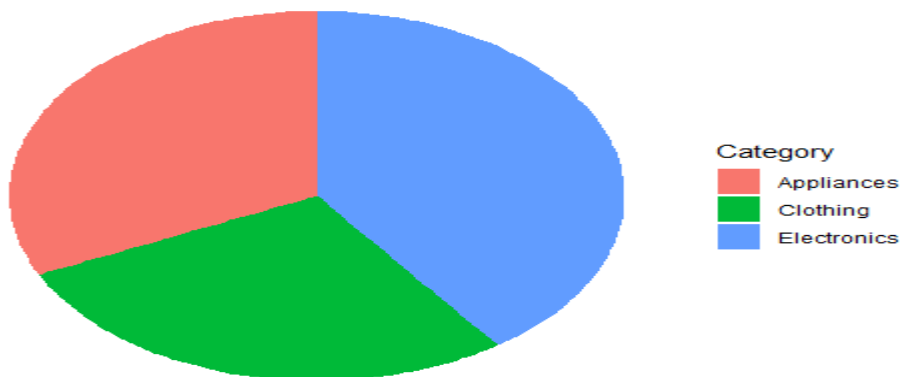
11 D. Pie Chart

Program:

```
ggplot(product_data,  
aes(x="",  
y=Sales,  
fill=Category))+  
geom_bar(stat="identity",width=1)+  
coord_polar("y")+  
labs(title="Sales Distribution Across Product Categories")+  
theme_void()
```

Output:

Sales Distribution Across Product Categories



SET 12- Website Traffic

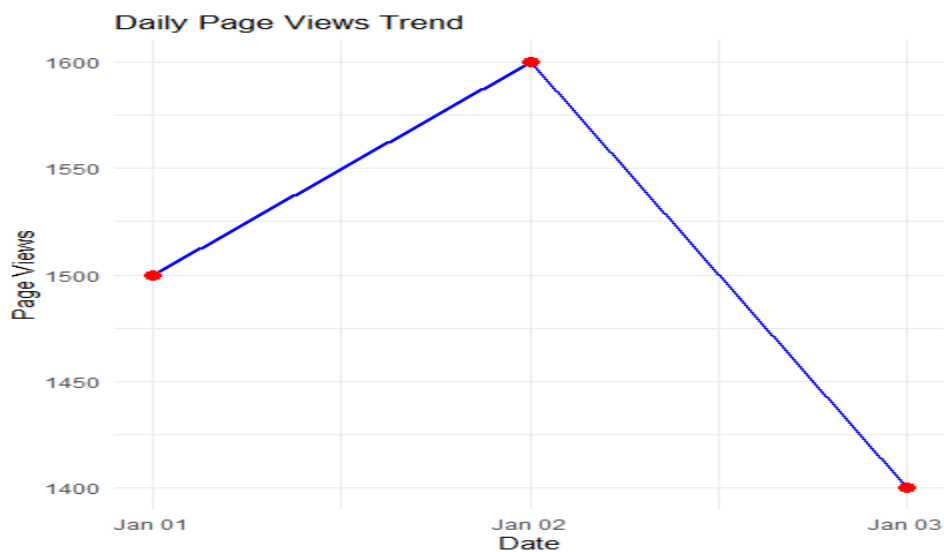
12 A. Line Chart

Program:

```
website_data <- data.frame(  
Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03")),  
PageViews=c(1500,1600,1400),  
CTR=c(2.3,2.7,2.0))  
ggplot(website_data,  
aes(x=Date,y=PageViews,group=1))+
```

```
geom_line(color="blue",size=1)+
geom_point(color="red",size=3)+
labs(title="Daily Page Views Trend",
x="Date",
y="Page Views")+
theme_minimal()
```

Output:

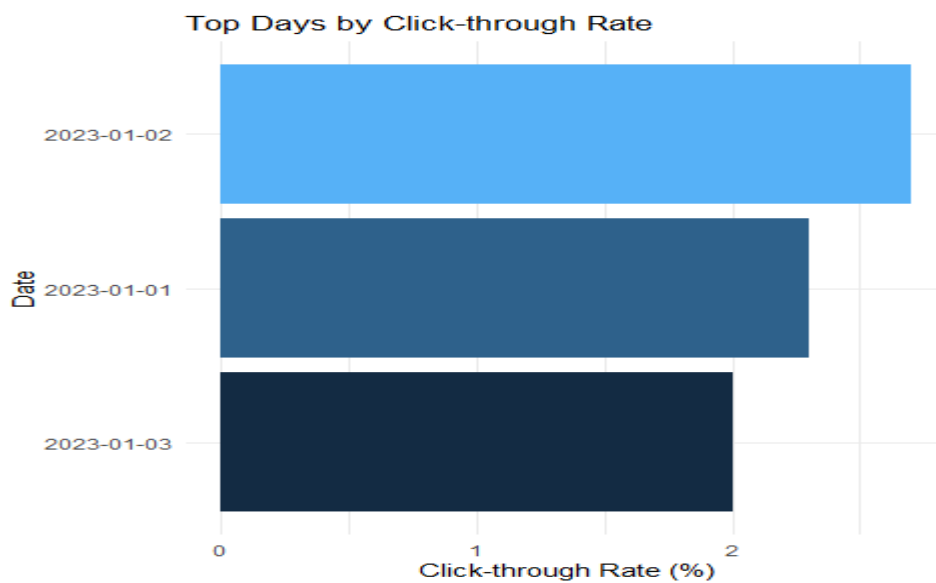


12 B. Bar Chart (Top Days by Click-through Rate)

Program:

```
top_ctr <- website_data[order(-website_data$CTR),]
ggplot(top_ctr,
aes(x=reorder(as.character(Date),CTR),
y=CTR,
fill=CTR))+
geom_bar(stat="identity")+
coord_flip()+
labs(title="Top Days by Click-through Rate",
x="Date",
y="Click-through Rate (%)")+
theme_minimal()+
theme(legend.position="none")
```

Output:



12 C. Table

Program:

```
kable(
  website_data,
  caption="Website Traffic Data")
```

Output:

```
Table: Website Traffic Data
|Date      | PageViews| CTR|
|:-----|:-----|:---|
|2023-01-01|      1500| 2.3|
|2023-01-02|      1600| 2.7|
|2023-01-03|      1400| 2.0|
> |
```

12 D. Interactive Dashboard

Program:

```
line_plot <- ggplotly(
  ggplot(website_data,
    aes(x=Date,y=PageViews,group=1))+
  geom_line(color="blue")+
  geom_point(color="red",size=3)+
  labs(title="Daily Page Views"))
```

```

bar_plot <- ggplotly(
  ggplot(top_ctr,
    aes(x=as.character(Date),
      y=CTR, fill=CTR))+
    geom_bar(stat="identity")+
    labs(title="Top CTR Days")+
    theme_minimal()+
    theme(legend.position="none"))
dashboard <- subplot(
  line_plot,
  bar_plot, nrow=1,
  shareX=FALSE,
  shareY=FALSE)
dashboard

```

Output:

